

①

* Example

$$\begin{aligned} a_{11}(t) &= 1 & a_{12}(t) &= 0 & b_{11}(t) &= -1 & b_{12}(t) &= -1 & f_1(t) &= 3t \\ a_{21}(t) &= 1 & a_{22}(t) &= 1 & b_{21}(t) &= -5 & b_{22}(t) &= -2 & f_2(t) &= 5 \end{aligned}$$

Then the systems of linear differential equations can be written in the form:

$$\begin{cases} x_1' + 0x_2' - x_1 - x_2 = 3t \\ x_1' + x_2' - 5x_1 - 2x_2 = 5 \end{cases} \Rightarrow \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} x_1' \\ x_2' \end{bmatrix}}_{\underline{x}'} + \underbrace{\begin{bmatrix} -1 & -1 \\ -5 & -2 \end{bmatrix}}_{\underline{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\underline{x}} = \underbrace{\begin{bmatrix} 3t \\ 5 \end{bmatrix}}_{\underline{f}(t)} \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}_{\underline{B}} \underline{x}' = \underbrace{\begin{bmatrix} +1 & +1 \\ +5 & +2 \end{bmatrix}}_{\underline{C}} \underline{x} + \underbrace{\underline{e}}_{\underline{f}} \Rightarrow \underline{B} \underline{x}' = \underline{C} \underline{x} + \underline{f}$$

$\Rightarrow$ Use Gauss-Jordan elimination to convert the system into the form of $\underline{x}' = \underline{A} \underline{x} + \underline{f}$

Let $\underline{E} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ The elementary row operation matrix that subtracts row 1 from row 2.

Then $\underline{E} \cdot \underline{B} = \underline{I} \quad \underline{A} = \underline{E} \underline{B} = \begin{bmatrix} +1 & +1 \\ +4 & +1 \end{bmatrix} \quad \underline{f} = \underline{E} \underline{f} = \begin{bmatrix} 3t \\ 5-3t \end{bmatrix}$

So the final system;

$$\underline{E} \underline{B} \underline{x}' = \underline{E} \underline{C} \underline{x} + \underline{E} \underline{f} \Rightarrow \underline{x}' = \underline{A} \underline{x} + \underline{f}$$

Solution: Use Laplace Transform

$$\begin{aligned} x_1' &= x_1 + x_2 + 3t \\ x_2' &= 4x_1 + x_2 + 5 - 3t \end{aligned} \Rightarrow \begin{aligned} sX_1(s) - x_1(0) &= X_1(s) + X_2(s) + \mathcal{L}\{3t\} \\ sX_2(s) - x_2(0) &= 4X_1(s) + X_2(s) + \mathcal{L}\{5-3t\} \end{aligned} \Rightarrow \boxed{s\underline{X}(s) - \underline{x}_0 = \underline{A}\underline{X}(s) + \underline{F}(s)}$$

$$(s\underline{I} - \underline{A})\underline{X}(s) = \underline{x}_0 + \underline{F}(s) \Rightarrow \boxed{\underline{X}(s) = (s\underline{I} - \underline{A})^{-1}\underline{x}_0 + (s\underline{I} - \underline{A})^{-1}\underline{F}(s)} *$$

(i) if we have zero initial conditions, then the solution only depends on the forcing functions (or inputs). $\underline{X}(s) = (s\underline{I} - \underline{A})^{-1}\underline{F}(s) \Rightarrow$ Zero state solution

(ii) if we have forcing functions being zero, then the solution only depends on the initial conditions (or states). $\underline{X}(s) = (s\underline{I} - \underline{A})^{-1}\underline{x}_0 \Rightarrow$ Homogeneous solution
Zero input solution

$$\underline{F}(s) = \begin{bmatrix} F_1(s) \\ F_2(s) \end{bmatrix} \quad F_1(s) = \int_0^\infty f_1(t) e^{-st} dt = \int_0^\infty 3t e^{-st} dt = \left(3t \cdot \frac{-1}{s} e^{-st} \right) \Big|_0^\infty - \int_0^\infty \left(-\frac{1}{s} e^{-st} \right) \cdot 3 dt = \frac{3}{s} \int_0^\infty e^{-st} dt$$

Integration by parts: $\int u dv = uv - \int v du \Rightarrow \begin{aligned} u &= 3t & dv &= e^{-st} dt \\ du &= 3 dt & v &= -\frac{1}{s} e^{-st} \end{aligned}$

$$F_2(s) = \int_0^\infty (5-3t) e^{-st} dt = \int_0^\infty 5e^{-st} dt - \frac{3}{s^2} = 5 \left(\frac{e^{-st}}{-s} \right) \Big|_0^\infty - \frac{3}{s^2} = \frac{5}{s} - \frac{3}{s^2}$$

$$= \frac{3}{s} \left(\frac{-1}{s} e^{-st} \right) \Big|_0^\infty = \frac{3}{s^2} \left(0 - \left(-\frac{1}{s} \right) \right)$$

②

$$* (sI - A)^{-1} = \begin{bmatrix} s-1 & -1 \\ -4 & s-1 \end{bmatrix}^{-1} = \frac{1}{s^2 - 2s - 3} \begin{bmatrix} s-1 & 1 \\ 4 & s-1 \end{bmatrix} \quad * \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$X(s) = (sI - A)^{-1} \underline{x}_0 + (sI - A)^{-1} \underline{F}(s) = \frac{1}{(s-3)(s+1)} \begin{bmatrix} s-1 & 1 \\ 4 & s-1 \end{bmatrix} \left(\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \begin{bmatrix} 3/s^2 \\ 5/s - 3/s^2 \end{bmatrix} \right)$$

$$= (sI - A)^{-1} (\underline{x}_0 + \underline{F}(s))$$

$$\frac{5}{s} - \frac{3}{s^2} = \frac{5s-3}{s^2}$$

$$= \frac{1}{(s-3)(s+1)} \begin{bmatrix} (s-1)x_1(0) + x_2(0) \\ 4x_1(0) + (s-1)x_2(0) \end{bmatrix} + \frac{1}{(s-3)(s+1)} \begin{bmatrix} \frac{3(s-1) + 5s-3}{s^2} \\ \frac{12 + (s-1)(5s-3)}{s^2} \end{bmatrix}$$

$$= \frac{1}{(s-3)} \underbrace{\begin{bmatrix} \frac{2x_1(0) + x_2(0)}{4} \\ \frac{2x_1(0) + x_2(0)}{2} \end{bmatrix}}_{\underline{c}_1} + \frac{1}{(s+1)} \underbrace{\begin{bmatrix} \frac{2x_1(0) - x_2(0)}{4} \\ \frac{x_2(0) - 2x_1(0)}{2} \end{bmatrix}}_{\underline{c}_2} + \underbrace{\frac{1}{s} \begin{bmatrix} -4 \\ 10/3 \end{bmatrix} + \frac{1}{s^2} \begin{bmatrix} 2 \\ -1 \end{bmatrix} + \frac{1}{s+3} \begin{bmatrix} 1/2 \\ 2/3 \end{bmatrix} + \frac{1}{s+1} \begin{bmatrix} 7/2 \\ -4 \end{bmatrix}}_{\underline{z}_s}$$

Zero Input - Homogeneous solution Zero-state solution

$$\xrightarrow{\mathcal{L}^{-1}} \underbrace{\underline{c}_1 e^{3t} + \underline{c}_2 e^{-t}}_{\underline{z}_1} + \underbrace{\underline{a}_1 + \underline{a}_2 t + \underline{a}_3 e^{3t} + \underline{a}_4 e^{-t}}_{\underline{z}_s} = \begin{bmatrix} \underline{c}_1 & \underline{c}_2 \end{bmatrix} \begin{bmatrix} e^{3t} \\ e^{-t} \end{bmatrix} + \begin{bmatrix} \underline{a}_1 & \underline{a}_2 & \underline{a}_3 & \underline{a}_4 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ e^{3t} \\ e^{-t} \end{bmatrix}$$

Forced solution.

⇒ this is an unstable system! $\lim_{t \rightarrow \infty} |e^{3t}| \rightarrow \infty$. * We have positive roots for the characteristic eqn. * Characteristic equation has mixed sign coefficients. $(s^2 - 2s - 3)$

→ Solution by State Transition matrix

* Let $\underline{\phi}(t; t_0, \underline{x}_0)$ be the solution to the homogeneous system ($\underline{x}' = \underline{A}\underline{x}$) with initial condition $\underline{x}_0$, e.g., From previous example $\underline{\phi}(t; t_0, \underline{x}_0) = \begin{bmatrix} \underline{c}_1 & \underline{c}_2 \end{bmatrix} \begin{bmatrix} e^{3t} \\ e^{-t} \end{bmatrix}$

(i) if $\underline{x}_0 = \sum_{i=1}^n \alpha_i \underline{x}_i$ then the solution to the homogeneous system with initial condition $\underline{x}_0$ is the linear combination of the solutions with initial conditions $\underline{x}_i$, $1 \leq i \leq n$, i.e., $\underline{\phi}(t; t_0, \underline{x}_0) = \sum \alpha_i \underline{\phi}(t; t_0, \underline{x}_i)$

Proof: $\underline{x}(t) = \underline{\phi}(t; t_0, \underline{x}_0) \quad \underline{x}' = \underline{A}\underline{x} \xrightarrow{\mathcal{L}^{-1}} (sI - \underline{A})\underline{X} = \underline{x}_0$

$$\underline{X}(s) = (sI - \underline{A})^{-1} \underline{x}_0$$

$$1 \leq i \leq n \quad \underline{\phi}'(t; t_0, \underline{x}_i) = \underline{A}\underline{\phi}(t; t_0, \underline{x}_i) \quad \uparrow \quad \sum \alpha_i \underline{\Phi}(s; t_0, \underline{x}_i) = (sI - \underline{A}) \sum \alpha_i \underline{x}_i$$

$$\downarrow \mathcal{L} \quad \underline{\Phi}(s; t_0, \underline{x}_i) = (sI - \underline{A})^{-1} \underline{x}_i \quad \uparrow \quad \sum \alpha_i \underline{\Phi}(s; t_0, \underline{x}_i) = \sum \alpha_i (sI - \underline{A})^{-1} \underline{x}_i$$

$$1 \leq i \leq n$$

③

* For n dimensional space we have at most n linearly independent vectors. Therefore we have only n LI solution to the homogeneous system, since the relation between the solutions is exactly same as the relation between the initial conditions, i.e., $\underline{x}_0 = \sum_{i=1}^n \alpha_i \underline{x}_i \rightarrow \underline{\phi}(t; t_0, \underline{x}_0) = \sum_{i=1}^n \alpha_i \underline{\phi}(t; t_0, \underline{x}_i)$

Example: From the previous example we know Laplace transform of the homogeneous solution. w/ $t_0=0$

$$\underline{\Phi}(s; t, \underline{x}_0) = \frac{1}{(s-3)} \begin{bmatrix} \frac{2x_1(0) + x_2(0)}{4} \\ \frac{2x_1(0) + x_2(0)}{2} \end{bmatrix} + \frac{1}{(s+1)} \begin{bmatrix} \frac{2x_1(0) - x_2(0)}{4} \\ \frac{x_2(0) - 2x_1(0)}{2} \end{bmatrix}$$

rearrange the $\underline{\Phi}(s; t, \underline{x}_0)$

$$= \begin{bmatrix} \frac{1}{2} \left(\frac{1}{s-3} + \frac{1}{s+1} \right) & \frac{1}{4} \left(\frac{1}{s-3} - \frac{1}{s+1} \right) \\ \frac{1}{s-3} - \frac{1}{s+1} & \frac{1}{2} \left(\frac{1}{s-3} + \frac{1}{s+1} \right) \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = (sI - A)^{-1} \underline{x}_0$$

$$\Rightarrow \underline{\phi}(t; t_0, \underline{x}_0) = \begin{bmatrix} \frac{1}{2} (e^{3t} + e^{-t}) & \frac{1}{4} (e^{3t} - e^{-t}) \\ e^{3t} - e^{-t} & \frac{1}{2} (e^{3t} + e^{-t}) \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} \frac{x_1(0)}{2} (e^{3t} + e^{-t}) + \frac{x_2(0)}{4} (e^{3t} - e^{-t}) \\ x_1(0) (e^{3t} - e^{-t}) + \frac{x_2(0)}{2} (e^{3t} + e^{-t}) \end{bmatrix}$$

$t_0=0$

* Fundamental matrix: Let $\underline{x}_i$ $1 \leq i \leq n$ be n LI vectors in $\mathbb{R}^n$. Then the solutions $\{ \underline{\phi}(t; t_0, \underline{x}_1), \dots, \underline{\phi}(t; t_0, \underline{x}_n) \}$ are LI solutions to the homogeneous system. $\underline{x}' = A\underline{x}$ and we call $\underline{\Phi}(t; t_0, \underline{x}_0)$ as the Fundamental Matrix.

where $\underline{\Phi}(t; t_0, \underline{x}_0) = [\underline{\phi}(t; t_0, \underline{x}_1) \quad \underline{\phi}(t; t_0, \underline{x}_2) \quad \dots \quad \underline{\phi}(t; t_0, \underline{x}_n)]$ and

$\underline{x}_0 = [\underline{x}_1 \quad \underline{x}_2 \quad \dots \quad \underline{x}_n]$. We can find solution of the homogeneous system with any initial condition $\underline{x}_0 \in \mathbb{R}^n$ by linear combination of the columns of the Fundamental matrix.

i.e. The initial condition $\underline{x}_0 = \sum_{i=1}^n \alpha_i \underline{x}_i = \underline{x}_0 \cdot \underline{\alpha}$ $\underline{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_n]^T$

then $\underline{\phi}(t; t_0, \underline{x}_0) = \sum_{i=1}^n \alpha_i \underline{\phi}(t; t_0, \underline{x}_i) = \underline{\Phi}(t; t_0, \underline{x}_i) \underline{\alpha}$

④

Example From previous example: $t_0 = 0$

$$\underline{a}(t) = \begin{bmatrix} \frac{1}{2}(e^{3t} - e^{-t}) \\ e^{3t} - e^{-t} \end{bmatrix}, \quad \underline{b}(t) = \begin{bmatrix} \frac{1}{4}(e^{3t} - e^{-t}) \\ \frac{1}{2}(e^{3t} + e^{-t}) \end{bmatrix}$$

Let $\underline{x}_0 = x_1(0) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2(0) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$, $\phi(t; t_0, \underline{x}_0) = \begin{bmatrix} \underline{a}(t) & \underline{b}(t) \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$

then the solutions due to initial conditions $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are $\underline{a}(t)$ and $\underline{b}(t)$ respectively

since $\underline{x}_0 = x_1(0) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2(0) \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \phi(t; t_0, \underline{x}_0) = x_1(0) \underline{a}(t) + x_2(0) \underline{b}(t)$. ✓

* Definition: State transition matrix

A special case of fundamental matrix: $\Phi(t; t_0, \underline{x}_0)$ is for $\underline{x}_0 = \underline{I}$ is called as the state transition matrix.

$$\Phi(t; t_0, \underline{I}) = [\phi(t; t_0, \underline{e}_1) \dots \phi(t; t_0, \underline{e}_n)] \text{ where } \underline{I} = [\underline{e}_1 \ \underline{e}_2 \dots \underline{e}_n]$$

$$\underline{x}(t) = \phi(t; t_0, \underline{x}_0) = \sum_{i=1}^n \alpha_i \phi(t; t_0, \underline{x}_i) = \Phi(t; t_0, \underline{x}_0) \underline{\alpha}$$

where $\underline{x}_0 = \sum \alpha_i \underline{x}_i = \underbrace{[\underline{x}_1 \dots \underline{x}_n]}_{\underline{x}_0} \underline{\alpha} \Rightarrow \underline{\alpha} = \underline{x}_0^{-1} \underline{x}_0$

$$\phi(t; t_0, \underline{x}_0) = \Phi(t; t_0, \underline{x}_0) \underline{x}_0^{-1} \underline{x}_0 = \underbrace{\Phi(t; t_0)}_{\text{State transition matrix}} \underline{x}_0 \Rightarrow \boxed{\Phi(t; t_0, \underline{x}_0) = \Phi(t; t_0) \underline{x}_0}$$

Example

$$\phi(t; t_0, \underline{x}_0) = \begin{bmatrix} \underline{a}(t) & \underline{b}(t) \end{bmatrix} \underline{x}_0 \Rightarrow \Phi(t; t_0=0) = \begin{bmatrix} \underline{a}(t) & \underline{b}(t) \end{bmatrix}$$

$$\Phi(t; t_0) = \mathcal{L}^{-1} \{ (sI - A)^{-1} \} = e^{At}$$

Find $\Phi(t; t_0)$ from Cayley-Hamilton theorem.

$$\underline{A} = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$$

$$\lambda_1 = -1$$

$$\lambda_2 = +3$$

$$e^{-t} = -\beta_1 + \beta_0$$

$$e^{3t} = 3\beta_1 + \beta_0$$

$$\beta_0 = \frac{1}{4}(e^{3t} + 3e^{-t})$$

$$\beta_1 = \frac{1}{4}(e^{3t} - e^{-t})$$

$$\Phi(t; t_0) = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \frac{1}{4}(e^{3t} - e^{-t}) + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{4}(e^{3t} + 3e^{-t})$$

$$= \frac{1}{4} \begin{bmatrix} 2(e^{3t} + e^{-t}) & (e^{3t} - e^{-t}) \\ 4(e^{3t} - e^{-t}) & 2(e^{3t} + e^{-t}) \end{bmatrix} \Rightarrow \text{same as the previous solution!}$$